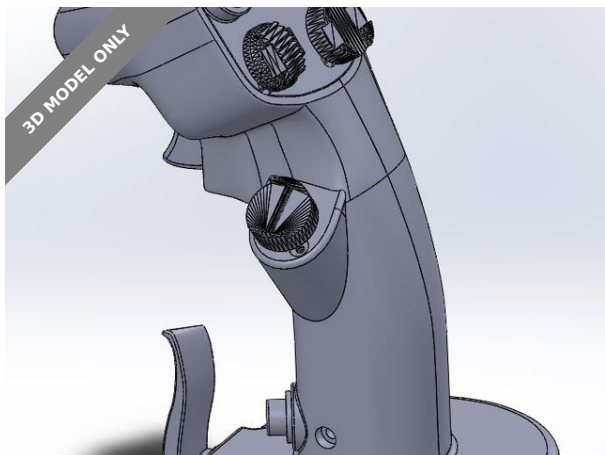




F-16 Sidestick Grip



Spock

[VIEW IN BROWSER](#)

updated 30. 5. 2024 | published 30. 5. 2024

Summary

This is a model of the F-16 sidestick controller designed to be used with flight simulators like DCS and Falcon BMS

[Gadgets](#) > [Video Games](#)

Tags: [joystick](#) [viper](#) [flightsim](#) [flightsimulation](#) [joystickmod](#)
[f16viper](#)

For older comments, [remixes](#), etc. see the [Thingiverse page](#) where I originally posted this model. Otherwise all files are posted exclusively here on Printables.

PRINTING

I printed the main left/right sides and the top face down with supports. I've included a zip file with Prusaslicer project files if you'd like to see where I added support blocking, etc. Most parts should be fairly self explanatory.

COMPONENTS

Printed

1x Left, Right, Top, Face Plate, Nubbin, Paddle, Trigger, DMS, TMS, CMS, Trim, Pickle, Pinkie, NWS, tact switch cover

4x hatswitch base, hatswitch cap, hatswitch stem

1-3x hatswitch push (optional, you can add a push button to the CMS, TMS, and Trim hats if desired)

Hardware

4x 5-10 mm M3 bolts (3 for securing Top, 1 for paddle)

2x 20-22 mm M3 bolts

10x 6-10 mm M2 screws/bolts (button head) for attaching hat switches and paddle

Can alternatively use 1x 26+ mm M2 bolt for paddle

2x Trigger spring (mine is 12mm long, 5mm OD, 4mm ID from a ballpoint pen)

1x Paddle spring (optional, >5mm ID, compressible to ~2mm)

22-25x 4.3x6x6 mm tact switches

May 2024

Cleaned up the BOM a little bit and moved it to the top of the page. Hopefully this is clearer and explains exactly what this is designed around. This design is pretty old now and honestly there are a lot of aspects that I am not a fan of. I have the reference material to rebuild it from the ground up (and I think I could make a much better model today than I did back in 2020) but unfortunately I don't have any access to SolidWorks right now so my CAD abilities are severely limited until I can learn and become proficient in some other CAD package. It's a pain to have to relearn all the tools but until then I probably won't be able to do anything too groundbreaking.

March 2024

It came to my attention that the folder containing the SolidWorks 2020 source files was messed up and most of the files were missing. This has been fixed and there is now a zip file containing the SolidWorks files too. In other news I'm also working on a redesign of this model from scratch. I want to better represent the way the switches etc are mounted irl, especially the trigger. I also think this one is about 5% too small in a few dimensions. I don't know how fast I'll be able to do this (I don't have SolidWorks access as of now) but stay tuned :)

July 2023

I have uploaded this to [printables.com](https://www.printables.com) because it's honestly just a much better website than Thingiverse. The Thingiverse page now points users

here.

[January 2024 Ok ok, Thingiverse isn't that bad anymore. But for a while I couldn't even log in or respond to messages. Files are still here because it's where I'm more active]

A few brief observations on the experiences of others and improvements I might try and make at some point in the future:

- The trigger mech is kinda finicky. Dependent on spring force and switch actuation force. Also has a bit of a bump to push through for the second stage. I have a new trigger design which I drew up sometime mid-2021 but it doesn't quite fit into the current model so I'm thinking this will require a full redraw of the stick. Honestly it kind of needs it.
- The screws are just M3 (or something around that size) direct into plastic, so not the best durability. In v2 I would probably try and add captive nuts or similar.
- The wiring, well, I actually haven't wired it, but that's because it is rather intimidating lol. I really like what Bacon8tor did with his remix [here](#) and might try to build off of his custom PCBs in future versions, making the whole setup a little more plug and play, fewer loose wires, etc. I also do really like the custom hat switches though, so we'll see.

Original Posting July 17 2020

Hello friends! This is a 3D printable F-16 (or A-10) grip, modelled by yours truly from a blender file and a bunch of reference photos of a Thrustmaster Warthog that a nice fella from the DIYHotas subreddit gave me ([check out his designs on thingiverse here](#)). I don't own a real F-16 SSC or a Warthog or Cougar so I can't speak as to how close this model really is, but I think it feels and looks about right.

I did not design this grip to attach to any specific joystick base. I left a 25mm hole on the base with a slot cut into the front and back and figured this might make it easy for people to make their own adapters for gimbals, extensions, whatever. Feel free to design directly into the grip if you wish. (also, would adding a screw hole down there for mounting be useful?)

I modeled this in two stages, first doing the external geometry and then slicing that up and clearing space for internals. I included two step files, one of the external stick geometry and another of the trigger.

My next project is going to be the F-16 TQS as I've already got a decent start on it. I began with hempstick's excellent shell model and going from there I've been adding electronics and a base. After that's done, who knows? I'm open to suggestions. I'm on the HOTAS Discord as Spock. If you think this model is worth it feel free to leave a tip, and of course let me

know if you have any problems with this design and I'll do my best to fix and improve.

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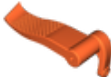
ASSEMBLY

The most finicky part of this design is the trigger. I've included a picture of the mechanism. Basically, place the tact switch cover over the tact switch, then the spring over that. If it's all good, you'll get the first stage click about halfway and it'll unclick when you release. If it won't release make sure the trigger moves quite freely all along the way. You may want to add a second spring (which I left room for above the first) to ensure the trigger always returns to the fully not-pressed state.

I think everything else is fairly self-explanatory. A lot of the tact switches for the various buttons need to have two of the prongs snapped off, and it's likely you'll need to cut a little material away on the NWS button hole in the right half to give enough space for the button. (and while we're on the subject, the NWS, Pickle, and Pinkie buttons are all slightly different lengths... I may fix that in the future but it's a bit too much math for me at the moment. Pinkie is the shortest and the wide part on the Pickle is thinner than the other two)

Oh, and the lower nubbin holds the tact switch for the pinkie button. The lower screw goes directly through it and into the right shell.

Model files

	STL Files	11 files
	tact-switch-cover.stl	
	viper-nubbin.stl	
	viper-nws.stl	
	viper-paddle-textured.stl	
	viper-pinkie.stl	



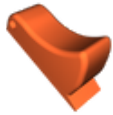
viper-pickle.stl



viper-face-plate.stl



viper-right.stl



viper-trigger.stl



viper-top.stl



viper-left.stl



STEP Files

5 files

viper-face-plate.step

viper-grip.step

viper-trigger-base.step

viper-grip-base.step

viper-stick-assem.step



alt-hatswitch-push.sldprt

alt-hatswitch-assem.sldasm

alt-hatswitch-base.sldprt

tact-switch-cover.sldprt

viper-trigger.sldprt

cms-hat-alt.sldprt

tact-switch-43x6x6.sldprt

alt-hatswitch-stem.sldprt

alt-hatswitch-cap.sldprt

dms-hat-alt.sldprt

tact-switch-5x6x6.sldprt

trim-hat.sldprt

viper-button-2.sldprt

tms-hat-alt.sldprt

paddle.sldprt

viper-stick-assem.sldasm

f-16-stick-no-tree-v2.sldprt

f-16-stick-no-tree-v2-left.sldprt



PrusaSlicer Files

4 files



viper-face-plate-ps.3mf



viper-top-ps.3mf



viper-left-ps.3mf



viper-right-ps.3mf

Other files

hat_switch_files_incl_step_and_sldprt.zip

left_handed_files.zip

solidworks-2020-files.zip

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